THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The electron has a: 1

(a) negative charge **(b)** double charge **(c)** positive charge **(d)** no charge

2. The K shell holds at most: 1

(a) 32 electrons **(b)** 2 electrons **(c)** 8 electrons **(d)** 18 electrons

3. Isotopes have the same: 1

(a) number of nucleons **(b)** number of neutrons
(c) mass number **(d)** atomic number

4. The proton has a: 1

(a) positive charge **(b)** no charge **(c)** zero mass **(d)** negative charge

5. The L shell holds at most: 1

(a) 18 electrons **(b)** 32 electrons **(c)** 8 electrons **(d)** 2 electrons

6. Isobars have the same: 1

(a) atomic number **(b)** number of protons
(c) mass number **(d)** valency

7. The neutron has: 1

(a) no mass **(b)** positive charge **(c)** no charge **(d)** negative charge

8. The electronic configuration of carbon is: 1

(a) 4, 2 **(b)** 2, 4 **(c)** 2, 2 **(d)** 2, 6

9. Hydrogen has: 1

(a) three isotopes **(b)** four isotopes **(c)** no isotope **(d)** two isotopes

10. The nucleus contains: 1

(a) protons and neutrons **(b)** only protons
(c) only neutrons **(d)** electrons only

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the sub-atomic particles with their charges. 2

12. What is valency? 2

13. Write the electronic configuration of sodium. 2

14. Who discovered the electron and the nucleus? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three observations of Rutherford's experiment. 3

16. Write the electronic configurations of Be, B, C. 3

17. List three uses of isotopes. 3

OR

Write three conclusions from Rutherford's experiment. 3

18. Write the electronic configurations of F, Ne, Na. 3

OR

State three limitations of Thomson's model. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe Thomson's model of the atom. 5

OR

Explain isotopes with examples and uses. 5

20. Describe Rutherford's model of the atom. 5

OR

Explain isobars with examples. 5

END OF QUESTION PAPER — Total: 40 Marks

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